Hence, $\rho(A) = \rho([A \mid B]) = 3$.

From the echelon form, we write the equivalent system of equations

$x + 2y - z = 3,\ 7y - 5z = 8,\ z = 4,\ 0 = 0.$

The last equation $0 = 0$ is meaningful. By the method of back substitution, we get

$$\begin{aligned} z &= 4 \\ 7y - 20 &= 8 \qquad \Rightarrow \quad y = 4, \\ x &= 3 - 8 + 4 \quad \Rightarrow \quad x = -1. \end{aligned}$$

So, the solution is $(x = -1, y = 4, z = 4)$. (Note that $A$ is not a square matrix.)

Here the given system is consistent and the solution is unique.

**Example 1.30**

Test for consistency of the following system of linear equations and if possible solve:

$4x - 2y + 6z = 8,\ x + y - 3z = -1,\ 15x - 3y + 9z = 21.$

**Solution**

Here the number of unknowns is 3.

The matrix form of the system is $AX = B$, where

$$A = \begin{bmatrix} 4 & -2 & 6 \\ 1 & 1 & -3 \\ 15 & -3 & 9 \end{bmatrix},\ X = \begin{bmatrix} x \\ y \\ z \end{bmatrix},\ B = \begin{bmatrix} 8 \\ -1 \\ 21 \end{bmatrix}.$$

Applying elementary row operations on the augmented matrix $[A \mid B]$, we get

$$[A \mid B] = \left[\begin{array}{ccc|c} 4 & -2 & 6 & 8 \\ 1 & 1 & -3 & -1 \\ 15 & -3 & 9 & 21 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 4 & -2 & 6 & 8 \\ 15 & -3 & 9 & 21 \end{array}\right] \xrightarrow{\substack{R_2 \to R_2 - 4R_1, \\ R_3 \to R_3 - 15R_1}} \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & -6 & 18 & 12 \\ 0 & -18 & 54 & 36 \end{array}\right]$$

$$\xrightarrow{\substack{R_2 \to R_2 \div (-6), \\ R_3 \to R_3 \div (-18)}} \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 1 & -3 & -2 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

So, $\rho(A) = \rho([A \mid B]) = 2 < 3$. From the echelon form, we get the equivalent equations

$$x + y - 3z = -1,\ y - 3z = -2,\ 0 = 0.$$

The equivalent system has two non-trivial equations and three unknowns. So, one of the unknowns should be fixed at our choice in order to get two equations for the other two unknowns. We fix $z$ arbitrarily as a real number $t$, and we get $y = 3t - 2$, $x = -1 - (3t - 2) + 3t = 1$. So, the solution is $(x = 1, y = 3t - 2, z = t)$, where $t$ is real. The above solution set is a one-parameter family of solutions. Here, the given system is consistent and has infinitely many solutions which form a one parameter family of solutions.

**Note**

In the above example, the square matrix $A$ is singular and so matrix inversion method cannot be applied to solve the system of equations. However, Gaussian elimination method is applicable and we are able to decide whether the system is consistent or not. The next example also confirms the supremacy of Gaussian elimination method over other methods.



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